Brandon Loaiza Rodriguez

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EDUCATION

Olin College of Engineering May 2030
B.S. in Mechanical Engineering GPA 4.0
Coursework: Applied Linear Algebra, Modeling & Simulation

College of Southern Nevada June 2026
Associate of Science GPA 3.65
Coursework: Multivariable Calculus, Statics

SKILLS

CAD & Simulation: SolidWorks (Surface Modeling, Assemblies), Onshape, SimScale CFD, Simcenter STAR-CCM+, Ansys
Fabrication: 3D Printing, Mill, Bandsaw, Hydraulic Press, TIG Welding, Composites, Injection Molding
Programming: MATLAB, Python
Automotive: OBD-II Diagnostics, Engine and Transmission Service, Brake and Suspension Repair, Cooling Systems
Languages: English (Fluent), Spanish (Native)

EXPERIENCE

Olin Electric Motorsports — Aerodynamics Team Member August 2026–Present

- Designed an undertray and diffuser in SolidWorks (surface modeling) and iterated through 24+ SimScale CFD simulations, raising downforce nearly 12x (6 N → 70 N) at 12 m/s with 14 N of drag. Writing aerodynamics documentation for the newly formed sub-team.
- Optimized angle of attack, throat cross-sectional area, and inlet strakes with rear vortex generators, using pressure distributions and velocity fields to guide each iteration.
- Work with the suspension team on ground-clearance targets and with the aero team on space claims and airflow requirements, including limiting diffuser exit flow that disrupts the side tunnels.

Loaiza Auto Repair — Apprentice Automotive Mechanic June 2020–October 2025

- Diagnosed engine-performance and drivability issues using OBD-II scan data, vehicle symptoms, and mechanical inspection.
- Worked alongside an experienced mechanic on engine and transmission removal/installation, wiring-harness reconnection, radiator and engine-mount replacement, and cooling-system service.
- Serviced and repaired 200+ vehicles across makes and models, working full-time summers and about 6 hours per week during the school year.

CSN Tutoring Center — College Math Tutor January 2026–July 2026

- Tutored about 20 students one-on-one in college math through Calculus II, 8 hours per week over 7 months; about 10 came back for repeat sessions.
- Used hands-on examples and step-by-step problem solving to explain challenging concepts.

PROJECTS

2006 Lexus IS350 — Engine, Drivetrain & Suspension Rebuild

- Replaced the engine with a family mechanic after tracing a coolant leak to a failed head gasket (about 10 days of hands-on work over 6 months), including harness reconnection, radiator and engine-mount replacement, and cooling-system bleeding.
- Traced a transmission stuck in 3rd gear to a failed solenoid box via fault codes; replaced the transmission and rear differential.
- Independently replaced worn shocks with coilovers and set ride height to fix a rough ride; traced abnormal vehicle movement after the repair to a severely deteriorated control-arm bushing.